

Relax. The test is less than one hour out of your long life. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points.

1. What is the importance of randomization when designing a scientific experiment?
 - A. Randomization is required to test between the null and alternative hypotheses.
 - B. Randomization is required to test between treatment and control groups.
 - C. Randomization is required to identify the independent and dependent variables. Independent variables are not randomized. Dependent variables are randomized.
 - D. Randomization is required to determine the proper significance level to use, such as $p \leq 0.05$.
 - E. Randomization is required to avoid introducing unwanted bias into the experimental results.
 2. If female birds are given a limited amount of food, then they will produce fewer eggs. This is an example of
 - A. a prediction of the experimental results that will support the tentative explanation for why female birds produce fewer eggs when food is scarce.
 - B. an observation that female birds produce few eggs when food is scarce.
 - C. a hypothesis because it is a tentative explanation for why female birds produce fewer eggs when food is scarce.
 - D. a hypothesis that explains the correlation between number of eggs produced and food availability.
 - E. a fact that the bird's diet includes compounds necessary to produce eggs.
 3. Which of the following statements *cannot* be tested by the scientific method?
 - A. The Grand Canyon was formed by a global flood about 4000 years ago.
 - B. All of these statements are religious claims so none can be tested by the scientific method.
 - C. The Earth is between 6000 and 10,000 years old.
 - D. God created light and dark, then stars.
 - E. The order in which organisms, including humans, first appeared on earth. For example, man, then plants, then land animals and birds, then woman.
 - F. The entire universe, the earth, and all living things were created in six days.
 4. After invention of the microscope, scientists discovered that all living organisms were composed of one or more cells, that the cell was the basic unit of function in all organisms, and that only existing cells gave rise to new cells. These specific observations were eventually organized into a general theory of biology called the Cell Theory. This is an example of
 - A. hypothetical reasoning.
 - B. objective reasoning.
 - C. scientific reasoning.
 - D. deductive reasoning.
 - E. inductive reasoning.
 - F. theoretical reasoning.
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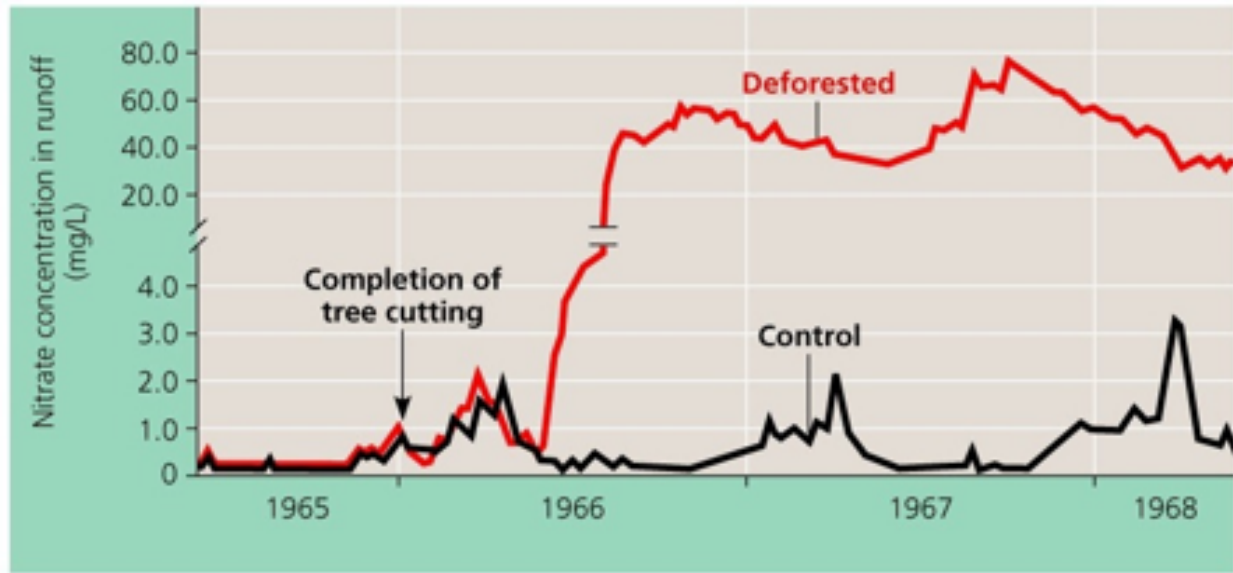
5. When you perform an experiment, the results are compared to the:
- A. Prediction
 - B. Hypothesis
 - C. Experiment
 - D. Conclusion
 - E. Observation
6. The Theory of Natural Selection, a basic theory of evolutionary biology, states that certain traits become more common in a population because the traits allow some individuals to produce more offspring. The Theory of Natural Selection is a theory because
- A. evolution can't be observed, so it can't be tested. It is basically a hypothesis.
 - B. it has been tested many times with many different hypotheses and has been proved to be true.
 - C. it is a tentative explanation of about how organisms adapt to their environment.
 - D. it has been tested many times with many different hypotheses and has always been supported.
 - E. it is not a fact, and may or may not be true.
7. Which of the following most accurately describes the process by which scientific knowledge advances?
- A. The use of inductive reasoning to explain specific experimental results about the natural world.
 - B. The use of deductive reasoning to develop broad scientific principles that explain the natural world.
 - C. The use of scientific analysis to explain the natural world by theory and experiments.
 - D. The analysis of the natural world through theoretical reasoning and experiments.
 - E. The objective analysis of the natural world by hypothesis testing and experiments.
8. You roll a pair of dice. What is the probability of getting a 2 and a 5, in any order? **Show your work. 3 points.**
- A. $1/3$
 - B. $1/10$
 - C. $1/36$
 - D. $1/18$
 - E. $1/2$
9. Prematurely born babies tend to have a higher death rate because they have not received all of the nutrients they need for proper development. This is an example of
- A. a theory that explains the well-known phenomenon of high death rate in prematurely born babies because it has been tested many times and has always been supported by the evidence.
 - B. an experiment that tests for death rate in prematurely born babies.
 - C. an observation that prematurely born babies experience high death rates.
 - D. a hypothesis because it is a tentative explanation for high death rate of prematurely born babies.
 - E. a fact that nutrition is important for fetal development.
10. Which of the following questions cannot be answered by the scientific method?
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- A. Can rats vomit?
 - B. Which fuel source is the most economical to replace coal?
 - C. Who deserves to be vaccinated in a time of crisis?
 - D. How far is the planet Neptune from the Sun?
 - E. What is the most nutritious food for cats?
11. Many ministers believe that praying for sick people will speed their recovery. This is
- A. not a scientific hypothesis because prayer is a supernatural phenomenon that can't be tested with an experiment.
 - B. a scientific hypothesis because the statement makes a specific claim that can be tested with an experiment.
 - C. a scientific hypothesis because sick people get better every day and people have prayed for them.
 - D. a scientific hypothesis because praying is a natural process that people do every day.
 - E. not a scientific hypothesis because the presence of a God that could answer prayer can't be tested with an experiment.
12. Standard deviation provides a measure of the amount of variation around the average value. About how much of the variation is contained within 1 standard deviation above and below ($\pm$) the average?
- A. 68%
 - B. 65%
 - C. 55%
 - D. 78%
 - E. 58%
13. A scientist predicted that fertilizer added to soil would cause plants to produce more fruit compared to soil that did not get fertilizer. After her experiment, plants that received fertilizer did produce more fruit than the other plants. Did she *prove* that fertilizer always causes plants to produce more fruit? (Choose the best answer.)
- A. Yes. The results of her experiment showed that adding fertilizer always causes plants to grow more fruit.
 - B. No. She showed that fertilizer caused those particular plants to produce more fruit. There is always a chance that other plants may not produce more fruit when given fertilizer.
 - C. Yes, because her hypothesis predicted that the plants would produce more fruit if fertilizer was added to the soil.
 - D. Maybe. She should do more experiments to be sure.
 - E. No. Her hypothesis was falsified because fertilizer causes plants to grow taller but doesn't affect fruit production.
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14. Why must an experiment include both treatment and control groups?

- A. To determine whether the results are significantly different from natural random variation.
- B. To determine the standard deviation around the average.
- C. To test both the null and alternative hypotheses.
- D. To include both independent and dependent variables in the experiment.
- E. To avoid introducing unwanted bias into the experiment.

15. The figure shows the amount of nitrogen contained in water runoff in a deforested area compared to a non-deforested, control area. Which of the following is the correct interpretation of this graph?



- A. The control experiment decreased the nitrogen concentration in the runoff.
- B. When trees are cut, they produce high amounts of nitrates that end up in the runoff.
- C. Tree cutting does not affect nitrate concentration because the deforested and control areas have the same concentration soon after tree cutting was complete.
- D. Deforestation increases the amount of runoff that contains nitrates.
- E. The nitrate concentration in runoff increased by as much as 8× higher in deforested areas compared to the control.

16. What is the probability of flipping a coin three times and getting three tails? **Show your work. 3 points.**

- A. 1/6
- B. 1/16
- C. 1/2
- D. 1/8
- E. 1/4

17. In science, a hypothesis is only useful if it
- A. can be tested.
 - B. can be proven correct.
 - C. agrees with the conclusion.
 - D. explains the natural world.
 - E. is proven correct.
18. When Abby Hodgins reported her experimental results, she included the probability value $p < 0.05$. This value indicates the probability that her results were due to her experiment treatment.
- A. True
 - B. False
19. Which of the following is the best definition of a hypothesis?
- A. A brief summary of a broad scientific theory.
 - B. An alternative explanation for a null hypothesis.
 - C. A tentative explanation for a natural phenomenon.
 - D. A prediction about the expected results from a specific experiment.
 - E. A tentative explanation about a supernatural phenomenon.

Read the following experiment to answer the next 10 questions about the scientific method and experimental design.

Honeybees are able to quickly find flowers for nectar and pollen. Early bee biologists did not know how bees located flowers. A famous bee biologist, Karl von Fritsche, thought that color may be important but he did not know if bees saw the world in color or in shades of grey. To find out, he trained two groups of bees to feed from a bowl of sugar water placed on a blue card. For the first group of bees, he placed the blue card in the middle of a set of grey cards. For the second group, he used only blue cards. No bowls were placed on any of the cards for either group. Von Fritsche thought that bees in the first group should always go to the blue card if they could see color but would go to any card if bees could not see color. He thought the bees in the second group should go to any card because the cards were all the same color. The bees in the first group always went to the blue card. The bees in the second group went to any card.

20. Identify the observation.
- A. Honeybees can see color.
 - B. Honeybees need nectar and pollen.
 - C. It was not known if honeybees could see color.
 - D. Honeybees can tell between blue and grey.
 - E. Honeybees can quickly find flowers.
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21. Identify the null hypothesis.

- A. Bees do not see color.
- B. Bees can be trained.
- C. Bees can see color.
- D. Bees do not see grey.
- E. Bees can see grey.

22. Identify the alternative hypothesis.

- A. Bees can be trained.
- B. Bees do not see grey.
- C. Bees do not see color.
- D. Bees can see grey.
- E. Bees can see color.

23. Identify the prediction made by the null hypothesis.

- A. If bees can see color then they will always go to the blue card.
- B. If bees can be trained then they will go to the blue card.
- C. If bees can see grey then they will go to the blue card.
- D. If bees cannot see grey then they will go to all of the cards.
- E. If bees cannot see color then they will go to the blue and grey cards.

24. Identify the prediction made by the alternative hypothesis.

- A. If bees can see color, then they will always go to the blue card.
- B. If bees can see grey, then they will go to the blue card.
- C. If bees cannot see color, then they will go to the blue and grey cards.
- D. If bees cannot see grey, then they will go to all of the cards.
- E. If bees can be trained, then they will go to the blue card.

25. Identify the dependent variable.

- A. The dependent variable was the trained bees.
 - B. The dependent variable was the response of the bees.
 - C. The dependent variable was the individual bees.
 - D. The dependent variable was the sugar water.
 - E. The dependent variable was the color of the cards.
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26. Identify the independent variable.

- A. The independent variable was the trained bees.
- B. The independent variable was the sugar water.
- C. The independent variable was the color of the cards.
- D. The independent variable was the response of the bees.
- E. The independent variable was the individual bees.

27. Identify the control group.

- A. The control group was the blue or grey cards.
- B. The control group was the bees that were given a choice of blue or grey cards.
- C. The control group was the bees that were trained to go to blue cards.
- D. The control group was the grey cards.
- E. The control group was the bees that were given only blue cards.

28. Identify the treatment group.

- A. The treatment group was the bees that were given a choice of blue or grey cards.
- B. The treatment group was the bees that were given only blue cards.
- C. The treatment group was the bees that were trained to go to blue cards.
- D. The treatment group was the blue or grey cards.
- E. The treatment group was the grey cards.

29. Identify the conclusion.

- A. Von Fritsche null hypothesis was supported. The bees see color instead of grey.
 - B. Von Fritsche's alternative hypothesis was supported. The bees see color instead of grey.
 - C. Von Fritsche null hypothesis was falsified. The bees see grey instead of color.
 - D. Von Fritsche's alternative hypothesis was falsified. The bees see grey instead of color.
 - E. Von Fritsche's alternative hypothesis was supported. The bees see grey instead of color.
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